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SYSTEM FOR TRANSFERRING ARTICLES FROM A CONTAINER

Abstract

An automated system for transferring articles from a container. An item transfer system includes a frame configured to receive a first tray and a second tray, a movable paddle assembly, a frame, and a plurality of actuators. The system is configured to transfer articles from the first tray to the second tray by an automated process including securing the articles within the first tray with the paddle assembly, moving a lifting assembly or an item support platform such that the articles are in a slidable configuration relative to the first tray, moving the paddle assembly and/or the item support platform in one or more linear translations to remove the items from the first tray and place the items above the second tray, and releasing the items into the second tray.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. application Ser. No. 17/810,571, filed Jul. 1, 2022, which is a divisional of U.S. application Ser. No. 16/870,605, filed May 8, 2020, titled “SYSTEM FOR TRANSFERRING ARTICLES FROM A CONTAINER,” which claims the benefit of U.S. Provisional Application No. 62/848,476, filed May 15, 2019, titled “SYSTEM FOR TRANSFERRING ARTICLES FROM A CONTAINER,” all of which are hereby incorporated by reference in their entirety.

FIELD

[0002] This disclosure relates to systems and methods for transfer of items between a first tray and a second tray.

DESCRIPTION OF THE RELATED TECHNOLOGY

[0003] Items, such as letters, envelopes, postcards, etc., may be received, transported, transferred, processed, collected, sorted, or the like, in containers such as trays. Items may be contained in different types of trays at various stages of transport and processing.

SUMMARY

[0004] The systems and methods of this disclosure each have several innovative aspects, no single one of which is solely responsible for its desirable attributes. Without limiting the scope as expressed by the claims that follow, its more prominent features will now be discussed briefly.

[0005] In one embodiment of the systems and methods disclosed herein, a tray content transfer system is described. The tray content transfer system comprises a frame comprising a first tray sub-frame coupled to the frame and configured to receive a first tray and a second tray sub-frame coupled to the frame and configured to receive a second tray at a location spaced from the first tray along a longitudinal axis; a paddle assembly movably coupled to the frame, the paddle assembly comprising a plurality of paddles, wherein the distance between the paddles is adjustable to selectively engage and release one or more items in the first tray, and wherein the paddle assembly is movable relative to the frame along a first axis parallel to the longitudinal axis and a second axis perpendicular to the first linear axis; a lifting assembly movably coupled to the frame, the lifting assembly comprising one or more lifting elements, wherein the lifting assembly is configured to lift the one or more items in the first tray by moving vertically along a third axis parallel to the second axis such that the one or more lifting elements pass at least partially through openings in a floor of the first tray; and an item support platform movably coupled to the frame, the item support platform comprising a substantially planar surface movable along a fourth axis parallel to the first axis between a first position above the first tray sub-frame and a second position above the second tray sub-frame.

[0006] In some embodiments, the tray content transfer system further comprises a controller configured to adjust the distance between the paddles, to move the paddle assembly along the first axis and the second axis, to move the lifting assembly along the third axis, and to move the item support platform along the fourth axis, in a predetermined sequence to transfer the one or more items from the first tray to the second tray. The predetermined sequence comprises: while the item support platform is not in the first position, causing the paddle assembly to engage the one or more items within the first tray; moving the paddle assembly upward along the second axis and moving the lifting assembly upward along the third axis to a raised position in which the items are disposed higher than a top of the first tray; moving the item support platform to the first position such that the item support platform is disposed between the one or more lifting elements and the items;

simultaneously moving the paddle assembly along the first axis and moving the item support platform along the fourth axis such that the item support platform is in the second position and the paddle assembly is disposed above the second tray; and moving the item support platform along the fourth axis away from the second position to release the one or more items into the second tray. In some embodiments, the tray content transfer system further comprises a sensor configured to detect the presence of the first tray or the second tray within the tray content transfer system, wherein the plurality of motors are configured to initiate the predetermined sequence based at least in part on detecting the presence of the first tray or the second tray.

[0007] In some embodiments, each paddle comprises a plurality of paddle fingers sized and shaped to enter one or more corresponding grooves within a sidewall of the first tray when the paddle assembly is lowered into the first tray. In some embodiments, the tray content transfer system further comprises one or more rails coupled to the frame parallel to the longitudinal axis at a height that retains the one or more items on the item support platform. In some embodiments, the tray content transfer system further comprises a conveyor coupled to the first tray sub-frame or the second tray sub-frame, the conveyor configured to move a first tray or a second tray along the frame parallel to the longitudinal axis. In some embodiments, the item support platform is further movable along the fourth linear axis to a disengaged position in which the item support platform is disposed above neither the first tray nor the second tray.

[0008] In another embodiment of the systems and methods disclosed herein, a tray content transfer system is described. The tray content transfer system comprises a frame comprising a first tray sub-frame coupled to the frame and configured to receive a first tray and a second tray sub-frame coupled to the frame and configured to receive a second tray at a location spaced from the first tray along a longitudinal axis, the second tray sub-frame disposed lower relative to the first tray sub-frame; a paddle assembly movably coupled to the frame, the paddle assembly comprising a plurality of paddles, a first paddle of the plurality of paddles configured to grasp the first sidewall and detach a removable sidewall from the first tray, wherein the distance between the paddles is adjustable to selectively engage and release one or more items in the first tray, and wherein the paddle assembly is movable relative to the frame along a first axis parallel to the longitudinal axis and a second axis perpendicular to the first axis; and an item support platform movably coupled to the frame, the item support platform comprising a substantially planar surface movable along a third axis parallel to the first axis between at least a first position above the second tray sub-frame and a second position in which the item support platform is not above the second tray sub-frame.

[0009] In some embodiments, the tray content transfer system further comprises a controller configured to adjust the distance between the paddles, to move the paddle assembly along the first axis and the second axis, and to move the item support platform along the third axis, in a predetermined sequence to transfer the one or more items from the first tray to the second tray. The predetermined sequence comprises moving the paddle assembly to engage the one or more items within the first tray; moving the first paddle to detach the removable sidewall from the first tray; moving the item support platform to the first position such that the item support platform is disposed above the second tray sub-frame and at least one of adjacent to or partially beneath the first tray proximate the first sidewall; moving the paddle assembly along the first axis to slide the one or more items from the first tray onto the item support platform, the one or more items retained between the detached removable sidewall and a second paddle of the plurality of paddles; and moving the item support platform along the fourth axis to the second position to release the one or more items into the second tray. In some embodiments, the tray content transfer system further comprises a sensor configured to detect the presence of the first tray or the second tray within the tray content transfer system, wherein the plurality of motors are configured to initiate the predetermined sequence based at least in part on detecting the presence of the first tray or the second tray.

[0010] In some embodiments, a second paddle of the plurality of paddles comprises a plurality of

paddle fingers sized and shaped to enter one or more corresponding grooves within a sidewall of the first tray when the paddle assembly is lowered into the first tray. In some embodiments, the tray content transfer system further comprises one or more rails disposed parallel to the longitudinal axis at a height that retains the one or more items on the item support platform. In some embodiments, the tray content transfer system further comprises a conveyor coupled to the first tray sub-frame or the second tray sub-frame, the conveyor configured to move a first tray or a second tray along the frame parallel to the longitudinal axis. In some embodiments, the item support platform is further movable along the fourth axis to a disengaged position in which the item support platform is disposed above neither the first tray nor the second tray.

[0011] In another embodiment of the systems and methods disclosed herein, a method for transferring items from a tray is described. The method comprises, by an automated process, receiving a first tray containing one or more items at a first location within a tray content transfer system; receiving a second tray within the tray content transfer system at a second location spaced along a longitudinal axis from the first location; securing the one or more items relative to the first tray; moving an item support platform parallel to the longitudinal axis and moving either a portion of the first tray or the one or more items, such that the one or more items are in a slidable configuration relative to the first tray; moving the one or more items along the longitudinal axis such that the one or more items rest on the item support platform above the second tray; and moving the item support platform parallel to the longitudinal axis to release the one or more items into the second tray.

[0012] In some embodiments, securing the one or more items comprises moving a paddle assembly downward into the first tray such that the one or more items are disposed between a plurality of paddles of the paddle assembly, and engaging the one or more items between the plurality of paddles, wherein the one or more items remain engaged between the plurality of paddles until the one or more items are released into the second tray. In some embodiments, moving either a portion of the first tray or the one or more items comprises detaching a detachable sidewall from the first tray. In some embodiments, moving either a portion of the first tray or the one or more items comprises lifting the one or more items out of the first tray by a lifting assembly. In some embodiments, the method is initiated based at least in part on a detection of the presence of the first tray or the second tray within the tray content transfer system. In some embodiments, the first tray is an automation-compatible tray and the second tray is not an automation-compatible tray.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The foregoing and other features of the disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings.

[0014] FIG. 1 is a schematic illustration of an example item processing system including a tray content transfer system.

[0015] FIG. 2A is a front perspective view of an example tray content transfer system.

[0016] FIG. 2B is a front elevation view of the tray content transfer system of FIG. 2A showing the axes of motion of the tray content transfer system.

[0017] FIG. 2C is a perspective view of an example source tray compatible with the tray content transfer system of FIGS. 2A and 2B.

[0018] FIG. 3 is a perspective view of an initial state of a tray content transfer system during an example tray content transfer process.

[0019] FIG. 4 is a perspective view of a second state of the tray content transfer system of FIG. 3 during the example tray content transfer process of FIG. 3.

[0020] FIG. 5 is a perspective view of a third state of the tray content transfer system of FIG. 3

during the example tray content transfer process of FIGS. 3-4.

[0021] FIG. 6 is a perspective view of a fourth state of the tray content transfer system of FIG. 3 during the example tray content transfer process of FIGS. 3-5.

[0022] FIG. 7 is a perspective view of a fifth state of the tray content transfer system of FIG. 3 during the example tray content transfer process of FIGS. 3-6.

[0023] FIG. 8 is a perspective view of a sixth state of the tray content transfer system of FIG. 3 during the example tray content transfer process of FIGS. 3-7.

[0024] FIG. 9 is a perspective view of a seventh state of the tray content transfer system of FIG. 3 during the example tray content transfer process of FIGS. 3-8.

[0025] FIG. 10A is a front perspective view of an example tray content transfer system.

[0026] FIG. 10B is a front elevation view of the tray content transfer system of FIG. 10A showing the axes of motion of the tray content transfer system.

[0027] FIG. 10C is a perspective view of an example source tray compatible with the tray content transfer system of FIGS. 10A and 10B.

[0028] FIG. 11 is a perspective view of an initial state of a tray content transfer system during an example tray content transfer process.

[0029] FIG. 12 is a perspective view of a second state of the tray content transfer system of FIG. 11 during the example tray content transfer process of FIG. 11.

[0030] FIG. 13 is a perspective view of a third state of the tray content transfer system of FIG. 11 during the example tray content transfer process of FIGS. 11-12.

[0031] FIG. 14 is a perspective view of a fourth state of the tray content transfer system of FIG. 11 during the example tray content transfer process of FIGS. 11-13.

[0032] FIG. 15 is a perspective view of a fifth state of the tray content transfer system of FIG. 11 during the example tray content transfer process of FIGS. 11-14.

[0033] FIG. 16 is a perspective view of a sixth state of the tray content transfer system of FIG. 11 during the example tray content transfer process of FIGS. 11-15.

[0034] FIG. 17 is a perspective view of a seventh state of the tray content transfer system of FIG. 11 during the example tray content transfer process of FIGS. 11-16.

[0035] FIG. 18 is a block diagram of a tray content transfer system.

[0036] FIG. 19 is a flowchart illustrating an example method of initiating the transfer of items from a container.

[0037] FIG. 20 is a flowchart illustrating an example method of transferring items from a source tray to an output tray in a tray content transfer system.

DETAILED DESCRIPTION

[0038] In the following detailed description, reference is made to the accompanying drawings. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. Thus, in some embodiments, part numbers may be used for similar components in multiple figures, or part numbers may vary from figure to figure. The illustrative embodiments described herein are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented. It will be readily understood that the aspects of the present disclosure and illustrated in the figures, can be arranged, substituted, combined, and designed in a wide variety of different configurations by a person of ordinary skill in the art, all of which are made part of this disclosure.

[0039] Reference in the specification to “one embodiment,” “an embodiment,” or “in some embodiments” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. Moreover, the appearance of these or similar phrases throughout the specification does not necessarily mean that these phrases all refer to the same embodiment, nor are separate or alternative embodiments necessarily mutually exclusive. Various features are described herein which may be exhibited by some embodiments and not by others. Similarly, various requirements are described which may be

requirements for some embodiments but may not be requirements for other embodiments.

[0040] In processing items in a distribution network, items can be transported in containers, such as trays. Items such as letters, postcards, envelopes, or other stackable items may be received, transported, and processed in trays. In some item processing implementations, two or more different types of trays may be used. When items are processed at a processing facility, a first type of tray may be optimized for use in automated processes and with automated processing equipment. For example, a first type of tray may be configured to be compatible with automated processing machinery at the processing facility (e.g., may have structural features designed to interact with particular processing machines and/or may be more rigid or dimensionally stable to facilitate automated processing of items). When the items have been processed and are ready to be transported outside of the facility, they may be placed into a second type of tray which may be flexible, lightweight, inexpensive to manufacture, sized and shaped to fit in an item collection system, or the like), such as on an aircraft. Thus, items received in the first type of tray may need to be transferred to the second type of tray after they are processed and before being transported away from the facility.

[0041] Transferring items between trays such as from rigid automation-friendly processing trays to soft-sided transport trays can be performed manually. Manual transfer of items from a first tray to a second tray can be time-consuming and imprecise. Items may be dropped while being manually transferred, for example, while an operator is moving a horizontally oriented stack of letters by exerting an inward pressure on the ends of the stack. In another example, items such as letters may be received in a uniformly faced arrangement, and operator error during transfer may cause some of the letters to face the opposite direction after transfer, resulting in processing errors or delays.

[0042] Some aspects of the tray content transfer systems disclosed herein are configured to provide an automated process for transferring items from a first tray to a second tray. In some embodiments, the tray content transfer systems may transfer items more reliably, efficiently, cost-effectively, and/or quickly than existing manual methods of tray content transfer. Although the present disclosure describes tray content transfer systems in the context of trays and flat items such as letter mail, it will be understood that other containers and items can be used without departing from the scope of the present disclosure.

[0043] Generally described, embodiments of the systems described herein use a sequence of motions to reliably and efficiently transfer items between a source tray, such as a rigid tray suited for use with machine processing of the items, and an output tray, such as a corrugated plastic, or letter tray, or other type of tray. For example, the items may be transferred from an automation-compatible tray to an output tray that is not an automation-compatible tray. In some embodiments, a loaded source tray and an empty output tray are placed within a tray content transfer system. A mail enclosure assembly descends to a position partially within the loaded source tray, such that the contents of the source tray are secured within the source tray. The tray content transfer system then moves the items and/or the source tray to a slidable configuration, such as by lifting the items and placing a platform or slide beneath the items, or by removing an end wall of the source tray. The items are then moved laterally to a position displaced vertically above the empty output tray. The slide or other platform supporting the items can be translated laterally while the items are laterally retained, to release the contents into the output tray. The transfer sequences disclosed herein can allow the transfer system to move articles such as letters, flats, parcels, and the like. In some embodiments, the system described herein can advantageously move items from a first tray to a second tray without lifting or suspending the articles over a space, gap, or area between the trays, where articles may be lost or damaged if dropped.

[0044] FIG. 1 schematically illustrates an example item processing system **100** including a tray content transfer system **105**. The item processing system **100** further includes a processing apparatus **110**, a transport location **120**, an empty source tray collection **130**, and an empty output tray supply **140**. Trays can be transferred between the components of the item processing system

100 by an intake flow path **115**, an output flow path **125**, an empty source tray flow path **135**, and an empty output tray flow path **145**. Each of the flow paths **115**, **125**, **135**, **145** can be a conveyor, such as a moving conveyor belt surface, a series of rollers, or any other conveying system configured to move one or more trays laterally and/or vertically.

[0045] The processing apparatus **110** can include one or more devices or systems for processing the items before tray content transfer. For example, the processing apparatus **110** can include any number of machines, such as mail processing equipment, configured to scan, image, weigh, measure, sort, order, combine, separate, analyze, or otherwise process the items. In some letter mail processing embodiments, the processing apparatus **110** can include one or more machines for automatically reading information provided on the letters (e.g., postage information, destination address, return address, etc.) and/or sorting the letters for further transportation according to item type and/or destination, or other criteria.

[0046] The transport location **120** can include one or more devices or systems for receiving item-containing output trays. For example, the transport location **120** can include an input of a loading dock, palletizer, container, etc. where item-containing trays are distributed to transport vehicles for transport away from the processing facility. In some embodiments, the transport location **120** can be an input of another process to be performed after tray content transfer. For example, in some letter mail processing embodiments, the transport location **120** may include a sorter/sequencer system configured to sort the letters into stop groups and sequence the letters into delivery sequence order.

[0047] The empty source tray collection **130** and empty output tray supply **140** can include one or more lines, piles, stacks, carts, dispensers, receivers, or other structures capable of holding a plurality of trays. For example, the empty source tray collection **130** may include a tray receiving system located at an end of the empty source tray flow path **135** for receiving and stacking or otherwise organizing empty source trays to be reused for transporting additional items. The empty output tray supply **140** can include one or more stacks, dispensers, or the like, for placing empty output trays into the empty output tray flow path **145** for delivery to the tray content transfer system **105**.

[0048] In an example method of operation, the tray content transfer system **105** receives an item-containing automation-compatible source tray from the processing apparatus **110** via the intake flow path **115**, and receives an empty output tray from the empty output tray supply **140** via the empty output tray flow path **145**. At the tray content transfer system **105**, the items in the source tray are transferred from the source tray to the empty output tray. After the transfer, the output tray contains the items previously received in the source tray. The item-containing output tray is then sent to the transport location **120** via the output flow path **125**. The empty source tray is sent to the empty source tray collection **130** via the empty source tray flow path **135**. After the source tray and output tray leave the tray content transfer system **105**, the example method can be repeated with a second item-containing source tray and a second empty output tray.

[0049] The example method described above with reference to FIG. **1** can be repeated any number of times or indefinitely, for example, based on a desired number or rate of item-containing source trays received for processing. In some embodiments, the tray content transfer system **105** can be configured to transfer the contents of two or more source trays to a single output tray, to transfer the contents of a single source tray to two or more output trays, or to transfer the contents of two or more source trays to two or more output trays simultaneously. In addition, various components of the item processing system **100** can be automated and/or performed manually. For example, any of the flow paths **115**, **125**, **135**, **145** can be performed manually at least in part, such as by an operator manually placing one or more loaded or empty trays onto a conveyor or directly into the tray content transfer system **105**.

[0050] FIGS. **2A** and **2B** depict a tray content transfer system **200** in accordance with an example embodiment. FIG. **2A** is a front perspective view of the tray content transfer system **200** in an

empty state. FIG. 2B is a front elevation view of the tray content transfer system **200** containing a source tray **280** and an output tray **290**, illustrating five axes of motion within the tray content transfer system **200**. FIG. 2C depicts an example source tray **280** compatible with the tray content transfer system **200**. The tray content transfer system **200** generally includes a frame **210**, a lifting assembly **220** movably mounted at least partially within the outer frame **210**, an item support platform **230**, and a paddle assembly **250** movably mounted above the frame **210**.

[0051] The frame **210** is generally configured to provide a stable support for the lifting assembly **220**, the item support platform **230**, and the paddle assembly **250**. The frame **210** includes various structural members **212** arranged to form a substantially rigid frame structure, upper frame members **214**, a source tray sub-frame **216**, an output tray sub-frame **218**, and one or more rails **211** positioned to laterally confine items being transferred. In some embodiments, the source tray sub-frame **216** includes one or more source tray conveyors **217** configured to slide a source tray **280** when the source tray **280** is at least partially on the source tray conveyors **217**. The output tray sub-frame **218** may similarly include one or more output tray conveyors **219** configured to slide an output tray **290** when the output tray **290** is at least partially on the output tray conveyors **219**. The source tray conveyors **217** and the output tray conveyors **219** may be reversible, for example, so that they can travel in a first direction before a transfer to pull a source tray **280** and/or an output tray **290** from an external conveyor system (e.g., an automated tray transport system or the like) into a suitable position within the tray content transfer system **200**, and can travel in a second direction after completing a transfer to push the source tray **280** and/or the output tray **290** back out to the external conveyor system. In various embodiments, the frame **210** can be any size, shape, or configuration suitable to support the various components of the tray content transfer system **200**. For example, the frame **210** can be built into a wall, a shelf, a floor, or the like, and need not be a free-standing rectangular frame as shown in the figures.

[0052] The upper frame members **214** of the frame **210** are generally configured to support and accommodate motion of a paddle assembly base **240** to which the paddle assembly **250** is mounted. A horizontal translation platform **242** is slidably mounted to a horizontal track member **244** disposed on the upper frame members **214**, to accommodate horizontal movement of the paddle assembly **250**. A vertical translation platform **246** is slidably mounted to two vertical track members **248** of the horizontal translation platform **242** to accommodate vertical movement of the paddle assembly **250**. Movement of the horizontal translation platform **242** along the horizontal track member **244** and movement of the vertical translation platform **246** along the vertical track members **248** can occur independently, and may be controlled by one or more motors, such as a horizontal paddle assembly motor **245** and a vertical paddle assembly motor **249**.

[0053] The lifting assembly **220** is configured to move vertically to lift items within a source tray **280**. Thus, the lifting assembly **220** includes an array of fingers **222** sized and shaped to fit within corresponding slots **282** in the floor of the source tray **280** (FIG. 2C). The lifting assembly **220** further includes vertical sliders **224** mounted on vertical tracks **228**. When the lifting assembly **220** is to be raised, the vertical sliders **224** can be driven upward along the vertical tracks **228**, for example, by one or more motors such as a lifting assembly motor **226**, applying a generally upward force on the horizontal sub-frame **222**. As the lifting assembly **220** travels upward, the tips **223** of the fingers **222** enter the slots **282**, passing through the floor of the source tray **280** to lift any items resting within the source tray **280**. The lifting assembly **220** may have a maximum range of vertical motion, for example, defined between the positions of the lifting assembly **220** when the vertical sliders **224** are at the bottom and top ends of the vertical tracks **228**.

[0054] The item support platform **230** is configured to move horizontally along a track **232** disposed above the source tray sub-frame **216** and the output tray sub-frame **218**. The track **232** may be mounted at an appropriate height such that the item support platform **230** can slide along the length of the track **232** unobstructed while a source tray **280** and an output tray **290** are located within the tray content transfer system **200**. For example, in some embodiments the item support

platform **230** is disposed at a predetermined height such that the item support platform **230** is slightly above the tips **223** of the fingers **222** (e.g., between $\frac{1}{8}$ inch and 1 inch, such as $\frac{1}{4}$ inch, $\frac{1}{2}$ inch, etc., above the tips **223**) when the lifting assembly **200** is in its highest position. The track **232** may provide a sufficient range of motion for the item support platform **230** such that the item support platform **230** can slide between a first extreme position (e.g., the disengaged position as shown in FIG. 2A in which the item support platform **230** is not above either the source tray sub-frame **216** or the output tray sub-frame **218**), and a second extreme position (e.g., the position as shown in FIG. 6 in which the item support platform **230** is above the source tray sub-frame **216**). In an intermediate position between the first and second extreme positions (e.g., the intermediate position shown in FIG. 7), the item support platform **230** is disposed above the output tray sub-frame **218**. In some embodiments, the item support platform **230** may comprise two or more pieces that are movable relative to each other. For example, the item support platform **230** may comprise two halves divided along a dimension of the item support platform **230** such that the item support platform **230** can open by rotating in a “bomb bay doors” configuration or by sliding outwardly.

[0055] The paddle assembly **250**, as shown in FIGS. 2A-2B, includes two paddles **252**, each paddle including a plurality of paddle fingers **254**. The paddle assembly **250** can be made of any substantially rigid material, for example, a metal such as sheet aluminum or steel, a hard plastic, or the like. One or more of the paddles **252** are slidably mounted on the paddle assembly base **240**. A paddle motor **255** may slide the paddles **252** to increase or decrease the distance between the paddles **252**.

[0056] Each paddle **252** is a generally planar surface disposed perpendicular to the paddle assembly base **240**. The paddles **252** include paddle fingers **254** spaced along the paddles **252** and generally extending downward from the paddles **252**. The paddle fingers **254** can be sized and shaped to fit within corresponding grooves between spacers **284** of a source tray **280**. As shown in FIG. 2C, the spacers **284** extend from the sidewalls of the source tray **280** to maintain a space between the sidewalls and any items within the source tray **280**. As will be described in greater detail with reference to FIGS. 3-9, the spacers **284** allow the paddle fingers **254** to slide downward into the space between each sidewall and any items in the source tray **280** such that the items can be contained between the paddles **252** and/or paddle fingers **254**. In various embodiments, the paddles **252** can have other sizes and/or shapes to conform with source trays **280** of any size or shape.

[0057] With reference to FIG. 2B, the components described above with reference to FIGS. 2A-2C can move along various axes with reference to the frame **210**. The paddles **252** of the paddle assembly **250** can be moved along several axes. Horizontal translation along a paddle assembly horizontal axis **241** can be achieved by moving the horizontal translation platform **242** along the horizontal track member **244** under control of the horizontal paddle assembly motor **245**. Vertical translation along a paddle assembly vertical axis **247** can be achieved by moving the vertical translation platform **246** along the vertical track members **248** under control of the vertical paddle assembly motor **249**. Squeezing of the paddles **252** along a paddle motion axis **253** can be achieved by moving one or both paddles **252** by the paddle motor **255**. Vertical translation of the lifting assembly **220** along a lifting assembly axis **225** can be achieved by moving the vertical sliders **224** along the vertical tracks **228** under control of the lifting assembly motor **226**. An item support motor (not shown) can move the item support **230** along an item support axis **231**. A combination of movements along axes **231**, **233**, **237**, **247**, and **253** can be used to transfer items from the source tray **280** to the process tray **290**.

[0058] Referring generally to FIGS. 2A and 2B, motion of any of the moving parts described herein, for example, horizontal translation along axes **231**, **241**, and **253**, and/or vertical translation along axes **225** and **247**, can be driven by any number of motors, electromechanical, pneumatic, etc., and can be selectively controlled and/or inhibited by mechanical or pneumatic brakes configured to stop and/or prevent motion along any of the movement axes described herein. The motors and/or brakes can be controlled by one or more controllers, which may include computer

components such as one or more processors, memory or other storage media, network or other communication interfaces, and/or other circuitry. In some embodiments, a memory of the controller stores computer-executable instructions that, when executed by the processor and/or other circuitry of the controller, directly or indirectly cause the motors and/or brakes to move in a predetermined sequence. The predetermined sequence can include, for example, a tray content transfer process such as the process described below with reference to FIGS. 3-9.

[0059] FIGS. 3-9 sequentially illustrate an example process for transferring the contents of a source tray **280** to an output tray **290**. Although the process of FIGS. 3-9 is shown and described in the context of the tray content transfer system **200** depicted in FIGS. 2A and 2B, it will be appreciated that the same or similar steps may be implemented in any other tray content transfer system.

[0060] FIG. 3 depicts an initial state of the tray content transfer process. In the initial state, a source tray **280** contains items **205**. The items can be, for example, letters, flats, etc. In the initial state of FIG. 3, the source tray **280** has been placed onto the source tray sub-frame **216** in a position below the paddle assembly **250**. An empty output tray **290** has been placed in the output tray sub-frame **218** spaced from the source tray **280** to receive the items **205** through the tray content transfer process. As described above with reference to FIG. 1, the source tray **280** and the output tray **290** may have been placed in their respective locations manually or automatically, for example, being placed into the tray content transfer system **200** by one or more conveyors, rollers, ramps, robotic arms, paddles, pushers, or other automated tray placement mechanism (e.g., by source tray conveyors **217** and/or output tray conveyors **219**), or manually by an operator. One or more alignment features may facilitate correct positioning of the source tray **280** and the output tray **290** within the tray content transfer system **200**. Alignment features may include, for example, one or more mechanical stops located on or adjacent to the source tray sub-frame **216** and/or the output tray sub-frame **218**, and/or may include source tray conveyors **217** and/or output tray conveyors **219** moving automatically to correctly position the source tray **280** and the output tray **290**. The paddle assembly **250** is located above the source tray **280** with the paddles **252** in an open configuration. The item support platform **230** is in a disengaged position away from the source tray sub-frame **216** and the output tray sub-frame **218** so as not to impede placement of the source tray **280** and the output tray **290**.

[0061] In an example tray content transfer process, the source tray **280** is an item-containing rigid plastic automation-friendly tray including various features optimized for transfer of the items **205** between various letter or flats processing machinery located elsewhere. The example output tray **290** is a corrugated plastic postal letter tray for the transportation of letter mail or flats. The items **205** stacked within the source tray **280** are to be transferred to the output tray **290**. In some embodiments, the items **205** may already be uniformly faced (that is, selectively flipped such that the postage and address of each item in the stack faces the same direction). Thus, it is desirable for the tray content transfer process to maintain the uniform facing of the items **205** during transfer.

[0062] After the item-containing source tray **280** and the empty output tray **290** are placed within the tray content transfer system **200**, the process continues to the configuration shown in FIG. 4. As shown in FIG. 4, the vertical translation platform **246** slides downward along the axis **247**, along the vertical track members **248** such that the paddle assembly **250** moves downward along the axis **247** to a location partially within the source tray **280**. The paddles **252** of the paddle assembly **250** are spaced such that the paddle fingers **254** of each paddle **252** descend into one of the empty spaces between the items **205** and an interior sidewall of the source tray **280** created by the spacers **284** (shown in FIG. 2C). The lower ends of the paddle fingers **254** can descend to a position at or near the bottom of the source tray **280** (e.g., within 1 inch, ½ inch, ¼ inch, or closer, or touching the bottom of the source tray **280**). Thus, in the state shown in FIG. 4, the stack of items **205** is disposed between the two paddles **252**. In some embodiments, one or both of the paddles **252** can be moved inward along axis **253** to compress the stack of items **205**, causing the items **205** to move to a more vertical orientation. In compressing the stack, the paddles **252** may push the items **205**

toward the middle of the source tray **280** and away from the ends, creating a space between the items **205** and the interior sidewalls of the source tray **280**. In various embodiments, the inward distance traveled by a paddle **252** can be, for example, $\frac{1}{2}$ inch, 1 inch, 2 inches, 3 inches, 4 inches, or more.

[0063] After the paddle assembly **250** is lowered into the source tray **280** and has compressed the stack of items **205**, the process continues to the configuration shown in FIG. 5. As shown in FIG. 5, the lifting assembly **220** and the paddle assembly **250** have moved upward in unison along the axes **225** and **247**, respectively. As the lifting assembly **220** moves upward, the fingers **222** enter the slots **282** in the bottom of the source tray **280**. The tips **223** of the fingers **222** impinge on the bottom surfaces of one or more of the items **205** in the source tray **280**, and push, force, urge, or otherwise elevate the items **205** upward such that the items **205** rest on the tips **223**. In some embodiments, the fingers **222** can move together to lift or move the stack of items **205** in the same plane. In some embodiments, the fingers **222** can move sequentially, at different rates, to different heights, and the like. For example, the fingers at one end may move to a higher point or different point than others of the fingers **222**. Although the fingers **222** include a plurality of separate members, the perpendicular orientation of the items **205** relative to the fingers **222** allows the tips **223** to serve as a suitable support for the items **205**. In some embodiments, the source tray **280** may further include a movable floor configured to be pushed upward by the fingers **222**, or by some other similar mechanism, such that the items **205** rest on the raised movable floor rather than directly on the tips **223** of the fingers **222**. In the state shown in FIG. 5, the items **205** are substantially confined between the paddles **252**, the tips **223** of the fingers **222** of the lifting assembly **220**, and the rails **211**, at a position higher than the sidewalls of the source tray **280** and the output tray **290**.

[0064] After the lifting assembly **220** and the paddle assembly **250** move upward to lift the stack of items **205** out of the source tray **280**, the process continues to the configuration shown in FIG. 6. As shown in FIG. 6, the item support platform **230** has been translated along axis **231** on the track **232** to a position above the source tray **280**. In translating to the position of FIG. 6, a leading edge of the item support platform **230** may pass between the tips **223** of the fingers **222** and the lower extremities of the paddle fingers **254**. Lower portions of the items **205** may deform slightly and/or items **205** may shift upward slightly to accommodate passage of the item support platform **230** between the finger tips **223** and the paddle fingers **254**. In the state shown in of FIG. 6, the items **205** now rest on the horizontally slidable item support platform **230**, rather than on the tips **223** of the fingers **222** which are only vertically translatable. Thus, the configuration of FIG. 6 is a slidable configuration that permits the items **205** to be moved to the output tray **290**. In some embodiments, the paddles **252** can compress the stack of items **205** and lift the items up a small amount to accommodate the passage of the item support platform **230** between the bottom surfaces of the items and the tips **223** of the fingers **222**. In some embodiments, the fingers **222** can sequentially retract a small amount to accommodate passage of the item support platform **230**. For example, the fingers **222** on the side closest to the item support platform **230** can retract, either fully, or a small amount, as the item support platform **230** nears or contacts those fingers **222**. As the item support platform **230** continues, the next finger **222** or set of fingers **222** can retract, to further allow passage of the item support platform **230**. This process can continue until the item support platform **230** is below the whole stack of items **205**.

[0065] After the items **205** are placed in the configuration shown in FIG. 6, the process continues to the configuration shown in FIG. 7. As shown in FIG. 7, the horizontal translation platform **242** has been translated along the axis **241**, along the horizontal track members **244**, to move the paddle assembly **250** to a position above the output tray **290**. At the same time, the item support platform **230** has been translated along the axis **231** to a position between the paddle assembly **250** and the output tray **290**. The paddle assembly **250** and the item support platform **230** can move in unison such that the items **250** remain supported by the item support platform **230** as the paddles **252**

move the items **205** laterally over the space, gap, or area between the source tray **280** and the output tray **290**. In some embodiments, one or more sensors may be provided to indicate when the paddle assembly **250** is positioned above the output tray **290**. This can prevent the items **205** from falling out of the transfer system and becoming damaged or lost.

[0066] After translation of the items **205** to the position above the output tray **290** shown in FIG. 7, the process continues to the configuration shown in FIG. 8. As shown in FIG. 8, the item support platform **230** has been translated back to its initial disengaged position to release the items **205** into the output tray **290**. In some embodiments, the item support platform **230** can instead return to a position above the source tray **280** to release the items **205**. In some embodiments, e.g., where the item support platform **230** comprises two or more separable pieces, the items **205** may be released into the output tray **290** by separating the pieces of the item support platform **230** in a sliding or bomb bay door opening motion. As the item support platform **230** retracts, friction between a top surface of the item support platform **230** and the bottom surfaces of one or more of the items can impart a force on one or more of the items, or on the entire stack. By moving downward, the paddles **252** continue to provide lateral support for the stack of items **205** against such a frictional force, preventing the items **205** from moving laterally with the item support platform **230**. The items fall down into the output tray **290**. In some embodiments, the paddles **252** can lower as the item support platform **230** retracts in order to ensure the items fall down into the output tray **290**, and do not fall or move too far laterally, missing the output tray **290**. In the configuration of FIG. 8, the items **205** have been transferred to the output tray **290** and are generally resting on the bottom interior surface of the output tray **290**. After the items **205** are released into the output tray **290**, the process continues to the configuration shown in FIG. 9.

[0067] Continuing to FIG. 9, the horizontal translation platform **242** returns to its initial state, and the now empty source tray **280** and the now occupied output tray **290** can be removed. The horizontal paddle assembly motor **245** and the vertical paddle assembly motor **249** move the paddle assembly **250** back to its initial position above the source tray **280** as shown in FIG. 3. The lifting assembly motor **226** moves the lifting assembly downward such that the fingers **222** are retracted from the slots **282** of the source tray **280**. Accordingly, the now-empty source tray **280** and the process tray **290** containing the items **205** can be removed from the tray content transfer system **200**, and the tray content transfer process can be repeated indefinitely as additional source trays **280** arrive with contents to be transferred.

[0068] FIGS. 10A and 10B depict a further example embodiment of a tray content transfer system **1000**. FIG. 10A is a front perspective view of the tray content transfer system **1000** in an empty state. FIG. 10B is a front elevation view of the tray content transfer system **1000** containing a source tray **1080** and an output tray **1090**, illustrating four axes of motion within the tray content transfer system **1000**. FIG. 10C depicts an example source tray **1080** compatible with the tray content transfer system **1000**. The tray content transfer system **1000** generally includes a frame **1010**, an item support platform **1030**, and a paddle assembly **1050** movably mounted above the frame **1010**.

[0069] The frame **1010** is generally configured to provide a stable support for the sliding platform **1030** and the paddle assembly **1050**. The frame **1010** includes various structural members **1012** arranged to form a substantially rigid frame structure, upper frame members **1014**, a source tray sub-frame **1016**, an output tray sub-frame **1018**, and one or more rails **1011** positioned to laterally confine items being transferred. In the example tray content transfer system **1010**, the output tray sub-frame **1018** is disposed at a lower height relative to the source tray sub-frame **1016**, such that the top of an output tray **1090** resting on the output tray sub-frame **1018** is at or below the level of the floor of a source tray **1080** resting on the source tray sub-frame **1016**. In some embodiments, the source tray sub-frame **1016** includes one or more source tray conveyors **1017** configured to slide a source tray **1080** when the source tray **1080** is at least partially on the source tray conveyors **1017**. The output tray sub-frame **1018** may similarly include one or more output tray conveyors

1019 configured to slide an output tray **1090** when the output tray **1090** is at least partially on the output tray conveyors **1019**. The source tray conveyors **1017** and the output tray conveyors **1019** may be reversible, for example, so that they can travel in a first direction before a transfer to pull a source tray **1080** and/or an output tray **1090** from an external conveyor system (e.g., an automated tray transport system or the like) into a suitable position within the tray content transfer system **1000**, and can travel in a second direction after completing a transfer to push the source tray **1080** and/or the output tray **1090** back out to the external conveyor system. In various embodiments, the frame **1010** can be any size, shape, or configuration suitable to support the various components of the tray content transfer system **1000**. For example, the frame **1010** can be built into a wall, a shelf, a floor, or the like, and need not be a free-standing rectangular frame as shown in the figures.

[0070] The upper frame members **1014** of the frame **1010** are generally configured to support and accommodate motion of a paddle assembly base **1040** to which the paddle assembly **1050** is mounted. A horizontal translation platform **1042** is slidably mounted to a horizontal track member **1044** disposed on the upper frame members **1014**, to accommodate horizontal movement of the paddle assembly **1050**. A vertical translation platform **1046** is slidably mounted to two vertical track members **1048** of the horizontal translation platform **1042** to accommodate vertical movement of the paddle assembly **1050**. Movement of the horizontal translation platform **1042** along the horizontal track member **1044** and movement of the vertical translation platform **1046** along the vertical track members **1048** can occur independently, and may be controlled by one or more motors, such as a horizontal paddle assembly motor **1045** and a vertical paddle assembly motor **1049**.

[0071] The item support platform **1030** is configured to move horizontally along a track **1032** disposed above the source tray sub-frame **1016** and the output tray sub-frame **1018**. The item support platform **1030** is mounted to the track **1032** by item support platform sliders **1036** and is movable along the track **1032** by an item support platform motor **1034**. The track **1032** may be mounted at an appropriate height such that the item support platform **1030** can slide along the length of the track **1032** unobstructed while an output tray **1090** is located within the tray content transfer system **1000**. For example, in some embodiments the item support platform **1030** is disposed at a predetermined height such that the item support platform **1030** is at approximately the level of the source tray sub-frame **1016**. The track **1032** may provide a sufficient range of motion for the item support platform **1030** such that the item support platform **1030** can slide between a first extreme position (e.g., the disengaged position as shown in FIG. **10A** in which the item support platform **1030** is not above the portion of the output tray sub-frame **1018** adjacent to the source tray sub-frame **1016**), and a second extreme position (e.g., the position as shown in FIG. **14** in which the item support platform **1030** is adjacent to the source tray sub-frame **1016**). In some embodiments, the item support platform **1030** may comprise two or more pieces that are movable relative to each other. For example, the item support platform **1030** may comprise two halves divided along a dimension of the item support platform **1030** such that the item support platform **1030** can open by rotating in a “bomb bay doors” configuration or by sliding outwardly.

[0072] The paddle assembly **1050**, as shown in FIGS. **10A-10B**, includes two paddles **1052a** and **1052b**. Paddle **1052a** includes grippers **1054a** configured to grasp, release, disengage, and/or otherwise actuate an end wall of a compatible source tray, as described below with reference to FIG. **10C**. Paddle **1052b** includes downwardly extending paddle fingers **1054b**. The paddle assembly **1050** can be made of any substantially rigid material, for example, a metal such as sheet aluminum or steel, a hard plastic, or the like. One or more of the paddles **1052a**, **1052b** are slidably mounted on the paddle assembly base **1040**. A paddle motor **1055** may slide the paddles **1052a**, **1052b** to increase or decrease the distance between the paddles **1052a**, **1052b**. Each paddle **1052a**, **1052b** is a generally planar surface disposed perpendicular to the paddle assembly base **1040**.

[0073] FIG. **10C** illustrates an example source tray **1080** compatible with the tray content transfer system **1000**. The source tray **1080** includes a detachable sidewall **1082**. The detachable sidewall

1082 may lock or latch to the source tray **1080**, and may be unlockable by engaging the grippers **1054a** within openings **1084** in a center portion of the top of the detachable sidewall **1082**. The paddle fingers **1054b** of paddle **1052b** can be sized and shaped to fit within corresponding grooves **1086** of the source tray **1080** which are disposed closer to the sidewalls of the source tray **1080** than the center of the source tray **180**, such that all items within the source tray **1080** can be contained between the paddle fingers **1054b** and the detachable sidewall **1082** when the paddle assembly **1050** is lowered into the source tray. In various embodiments, the paddles **1052a**, **1052b** can have other sizes and/or shapes to conform with source trays **1080** of any size or shape.

[0074] With reference to FIG. **10B**, the components described above with reference to FIGS. **10A-10C** can move along various axes with reference to the frame **1010**. The paddles **1052a**, **1052b** of the paddle assembly **1050** can be moved along several axes. Horizontal translation along a paddle assembly horizontal axis **1041** can be achieved by moving the horizontal translation platform **1042** along the horizontal track member **1044** under control of the horizontal paddle assembly motor **1045**. Vertical translation along a paddle assembly vertical axis **1047** can be achieved by moving the vertical translation platform **1046** along the vertical track members **1048** under control of the vertical paddle assembly motor **1049**. Squeezing of the paddles **1052a**, **1052b** along a paddle motion axis **1053** can be achieved by moving one or both paddles **1052a**, **1052b** by the paddle motor **1055**. The item support platform motor **1034** can move the item support **1030** along an item support axis **1031**. A combination of movements along axes **1031**, **1041**, **1047**, and **1053** can be used to transfer items from the source tray **1080** to the process tray **1090**.

[0075] Referring generally to FIGS. **10A** and **10B**, motion of any of the moving parts described herein, for example, horizontal translation along axes **1031**, **1041**, and **1053**, and/or vertical translation along axis **1047**, can be driven by any number of motors, and can be selectively controlled and/or inhibited by mechanical brakes configured to stop and/or prevent motion along any of the movement axes described herein. The motors and/or brakes can be controlled by one or more controllers, which may include computer components such as one or more processors, memory or other storage media, network or other communication interfaces, and/or other circuitry. In some embodiments, a memory of the controller stores computer-executable instructions that, when executed by the processor and/or other circuitry of the controller, directly or indirectly cause the motors and/or brakes to move in a predetermined sequence. The predetermined sequence can include, for example, a tray content transfer process such as the process described below with reference to FIGS. **11-17**.

[0076] FIGS. **11-17** sequentially illustrate an example process for transferring the contents of a source tray **1080** to an output tray **1090**. Although the process of FIGS. **11-17** is shown and described in the context of the tray content transfer system **1000** depicted in FIGS. **10A** and **10B**, it will be appreciated that the same or similar steps may be implemented in any other tray content transfer system.

[0077] FIG. **11** depicts an initial state of the tray content transfer process. In the initial state, a source tray **1080** contains items **1005**. The items can be, for example, letters, flats, etc. In the initial state of FIG. **11**, the source tray **1080** has been placed onto the source tray sub-frame **1016** in a position below the paddle assembly **1050**. An empty output tray **1090** has been placed in the output tray sub-frame **1018** spaced from and lower than the source tray **1080** to receive the items **1005** through the tray content transfer process. As described above with reference to FIG. **11**, the source tray **1080** and the output tray **1090** may have been placed in their respective locations manually or automatically, for example, being placed into the tray content transfer system **1000** by one or more conveyors, rollers, ramps, robotic arms, paddles, pushers, or other automated tray placement mechanism (e.g., by source tray conveyors **1017** and/or output tray conveyors **1019**). One or more alignment features may facilitate correct positioning of the source tray **1080** and the output tray **1090** within the tray content transfer system **1000**. Alignment features may include, for example, one or more mechanical stops located on or adjacent to the source tray sub-frame **1016** and/or the

output tray sub-frame **1018**, and/or may include source tray conveyors **1017** and/or output tray conveyors **1019** moving automatically to correctly position the source tray **1080** and the output tray **1090**. The paddle assembly **1050** is located above the source tray **1080** with the paddles **1052a**, **1052b** in an open configuration. The item support platform **1030** is in a disengaged position away from the source tray sub-frame **1016** so as not to impede placement of the source tray **1080** and the output tray **1090**.

[0078] In the example tray content transfer process, the source tray **1080** is a rigid plastic automation-friendly or automation-compatible tray including various features optimized for transfer of the items **1005** between various letter or flats processing machinery located elsewhere. The example output tray **1090** is a corrugated plastic postal letter tray for the transportation of letter mail or flats. The items **1005** stacked within the source tray **1080** are to be transferred to the output tray **1090**. In some embodiments, the items **1005** may already be uniformly faced (that is, selectively flipped such that the postage and address of each item in the stack faces the same direction). Thus, it is desirable for the tray content transfer process to maintain the uniform facing of the items **1005** during transfer.

[0079] After the source tray **1080** and the output tray **1090** are placed within the tray content transfer system **1000**, the process continues to the configuration shown in FIG. 12. As shown in FIG. 12, the vertical translation platform **1046** slides downward along the axis **1047**, along the vertical track members **1048** such that the paddle assembly **1050** moves downward along the axis **1047** to a location partially within the source tray **1080**. The paddles **1052a**, **1052b** of the paddle assembly **1050** are spaced such that the grippers **1054a** of paddle **1052a** descend into the openings **1084** in the top of the detachable sidewall **1082** and the paddle fingers **1054b** descend into the grooves **1086** (shown in FIG. 10C). The lower ends of the paddle fingers **1054b** can descend to a position at or near the bottom of the source tray **1080** (e.g., within 1 inch, ½ inch, ¼ inch, or closer, or touching the bottom of the source tray **1080**). Thus, in the state shown in FIG. 12, the stack of items **1005** is disposed between the paddle fingers **1054b** and the detachable sidewall **1082**. In some embodiments, paddle **1052b** can be moved inward along axis **1053** to compress the stack of items **1005**, causing the items **1005** to move to a more vertical orientation. In various embodiments, the inward distance traveled by paddle **1052b** can be, for example, ½ inch, 1 inch, 2 inches, 3 inches, 4 inches, or more.

[0080] After the paddle assembly **1050** is lowered into the source tray **1080**, the process continues to the configuration shown in FIG. 13. As shown in FIG. 13, the grippers **1054a** have detached the detachable sidewall **1082**, and the paddle assembly **1050** has traveled upward to lift the detachable sidewall **1082** above the floor of the source tray **1080**. The upward distance traveled by the paddle assembly **1050** may be, for example, approximately ¼ inch, ½ inch, 1 inch, or more. In the state shown in FIG. 13, the items **1005** are substantially confined between the paddle fingers **1054b**, the floor of the source tray **1080**, and the detachable sidewall **1082** of the source tray **1080**.

[0081] After the paddle assembly **1050** moves upward to partially lift the detachable sidewall **1082** of the source tray **1080**, the process continues to the configuration shown in FIG. 14. As shown in FIG. 14, the item support platform **1030** has been translated along axis **1031** by the item support platform motor **1034** to a position above the output tray **1090**. In translating to the position of FIG. 14, a leading edge of the item support platform **1030** travels to a position adjacent to or slightly beneath an end of the source tray **1080**. The item support platform **1030** at least partially covers the output tray **1090** and provides a surface nearer to the height of the floor of the source tray **1080** such that the stack of items **1005** can slide onto the item support platform **1030** without falling into the output tray **1090**. Thus, the configuration of FIG. 14 is a slidable configuration that permits the items **1005** to be moved to the output tray **1090**.

[0082] After the items **1005** are placed in the slidable configuration of FIG. 14, the process continues to the configuration shown in FIG. 15. As shown in FIG. 15, the horizontal translation platform **1042** has been translated along the axis **1041**, along the horizontal track members **1044**, to

move the paddle assembly **1050** to a position above the output tray **1090**. As the paddle assembly **1050** moves, the items **1005** contained within the paddle assembly **1050** slide in a stack onto the item support platform **1030**. Due to the presence of the item support platform **1030**, any downward movement of the items **1005** is not sufficient to cause any of the items **1005** to fall out from between the detachable sidewall **1082** and the paddle fingers **1054b**. Thus, the items **1005** in FIG. **15** are retained in a position above the output tray **1090**.

[0083] After translation of the items **1005** to a position above the output tray **1090**, the process continues to the configuration shown in FIG. **16**. As shown in FIG. **16**, the item support platform **1030** has been translated back to its initial disengaged position to release the items **1005** into the output tray **1090**. In some embodiments, e.g., where the item support platform **1030** comprises two or more separable pieces, the items **1005** may be released into the output tray **1090** by separating the pieces of the item support platform **1030** in a sliding or bomb bay door opening motion. In the configuration of FIG. **16**, the items **1005** have been transferred to the output tray **1090** and are generally resting on the bottom interior surface of the output tray **1090**. After the items **1005** are released into the output tray **1090**, the process continues to the configuration shown in FIG. **17**.

[0084] Continuing to FIG. **17**, the paddle assembly **1050** returns to its initial state such that the now-empty source tray **1080** and the now-occupied output tray **1090** can be removed, either automatically or manually. The horizontal paddle assembly motor **1045** and the vertical paddle assembly motor **1049** move the paddle assembly **1050** back to its initial position above the source tray **1080** as shown in FIG. **11**. The detachable sidewall **1082** is released from the grippers **1054a** and can be placed back into its initial position within the source tray **1080**. Accordingly, the now-empty source tray **1080** and the process tray **1090** containing the items **1005** can be removed from the tray content transfer system **1000**, and the tray content transfer process can be repeated indefinitely as additional source trays **1080** arrive with contents to be transferred.

[0085] FIG. **18** is a block diagram of a tray content transfer system **1800** in accordance with an example embodiment. The system **1800** includes a sensor **1810** in communication with a CPU **1820**. The CPU **1820** is further in communication with a memory **1830** and actuators including actuator **1840**, actuator **1850**, actuator **1860**, actuator **1870**, and actuator **1880**. Although the system **1800** is depicted as having 5 actuators, the number of actuators in various embodiments can be greater or smaller than 5, for example, based on the number of parts to be moved in any particular embodiment of the system **1800**.

[0086] The sensor **1810** is configured to detect the presence of one or more containers or other objects within the system **1800**. For example, the sensor **1810** can be configured to detect the presence of a source tray and/or an output tray before initiating a transfer process. In various implementations, the sensor **1810** can include one or more proximity sensors, profile sensors, pressure sensors, force sensors, scales, cameras, optical pattern recognition devices such as barcode scanners, light gates, or the like. In some embodiments, the sensor **1810** can include a manually activated device, such as a button, configured to produce a signal indicating that a source tray and an output tray have been loaded into the system **1800**. In some embodiments, the sensor **1810** can include a first sensor configured to detect the presence of a source tray and a second sensor configured to detect the presence of a process tray.

[0087] The CPU **1820** is configured to control and receive input from the sensor **1810**, the memory **1830**, and the actuators **1840**, **1850**, **1860**, **1870**, **1880**. The memory **1830** can store data received from the CPU **1820** and send data stored therein to the CPU **1820**. Examples of information that may be received and stored in the memory **1830** include, for example, information received at the CPU **1820** from the sensor **1810**, information received at the CPU from the actuators **1840**, **1850**, **1860**, **1870**, **1880**, and one or more computer-executable instructions that, when executed by the CPU **1820**, cause the CPU to selectively activate and/or deactivate the actuators **1840**, **1850**, **1860**, **1870**, **1880**, **1890** in a predetermined sequence to transfer items from a source tray to an output tray.

[0088] The actuators **1840**, **1850**, **1860**, **1870**, **1880** can be electronically controllable actuators each coupled to and configured to move one or more mechanical components of the system **1800**. Each actuator **1840**, **1850**, **1860**, **1870**, **1880** can include one or more electric motors, hydraulic cylinders, pneumatic actuators, screw jacks, servos, solenoids, or the like. In the example tray content transfer system **200** depicted in FIGS. 2A-2B, the actuators may be arranged such that actuator **1840** corresponds to the lifting assembly motor **226**, actuator **1850** corresponds to the horizontal paddle assembly motor **245**, actuator **1860** corresponds to the vertical paddle assembly motor **249**, actuator **1870** corresponds to the paddle motor **255**, and actuator **1880** is configured to slide the item support platform **230** along axis **231** on the track members **231**. In the example tray content transfer system **1000** depicted in FIGS. 10A and 10B, the actuators may be arranged such that actuator **1840** corresponds to the horizontal paddle assembly motor **1045**, actuator **1850** corresponds to the vertical paddle assembly motor **1049**, actuator **1860** corresponds to the paddle motor **1055**, actuator **1870** corresponds to the item support platform motor **1034**, and actuator **1880** is configured to actuate the grippers **1054a** to grasp and release the detachable sidewall **1082** of the source tray **1080**.

[0089] FIG. 19 is a flowchart illustrating an example method **1900** of initiating the transfer of items from a container. The method **1900** can be performed by a computer system integrated within a system such as the tray content transfer systems **105**, **200**, **1000**, **1800** depicted in FIGS. 2A-18. Although the method **1900** is described with reference to the tray content transfer system **1800**, it may equally be performed within any of the tray content transfer systems described herein.

[0090] The method **1900** begins with block **1910**, where the system **1800** begins receiving information from the sensor **1810**. The information can be an electrical signal indicating that a source tray and/or an output tray have or have not been detected within the system **1800**. When the system **1800** has begun receiving information from the sensor **1810**, the method **1900** continues to decision state **1920**.

[0091] At decision state **1920**, it is determined whether a source tray has been detected. For example, the CPU **1820** can analyze a signal received from the sensor **1810** configured to detect the presence of a source tray within the system **1800** at a suitable location for transferring items from the source tray. If it is determined that a source tray has not been detected, the method **1900** continues to block **1925**, where the system **1800** waits for a source tray to be detected and returns to decision state **1920**. If it is determined that a source tray has been detected, the method **1900** continues to decision state **1930**.

[0092] At decision state **1930**, it is determined whether an output tray has been detected. For example, the CPU **1820** can analyze a signal received from the sensor **1810** configured to detect the presence of an output tray within the system **1800** at a suitable location for receiving items transferred from the source tray. If it is determined that an output tray has not been detected, the method **1900** continues to block **1935**, where the system **1800** waits for an output tray to be detected and returns to decision state **1930**. If it is determined that an output tray has been detected, the method **1900** continues to block **1940**. At block **1940**, the method **1900** terminates by initiating the transfer of items from the source tray to the output tray.

[0093] FIG. 20 is a flowchart illustrating an example method **2000** of transferring items from a source tray to an output tray in a tray content transfer system. The method **2000** can be initiated at block **1940** of the method **1900** depicted in FIG. 19. The method **2000** can be performed by a computer integrated within a tray content transfer system such as the systems **200**, **1000**, and **1800** depicted in FIGS. 2A-18.

[0094] The method **2000** begins at block **2002**, where the source tray **280**, **1080** and output tray **290**, **1090** are received within the tray content transfer system **200**, **1000**. Receiving the source tray **280**, **1080** and the output tray **290**, **1090** can include an operator placing the source tray **280**, **1080** and/or the output tray **290**, **1090**. Receiving the source tray **280**, **1080** and the output tray **290**, **1090** may also include moving the source tray **280**, **1080** and/or the output tray **290**, **1090** into position

on respective sub-frames by conveyors 217, 219, 1017, and/or 1019. After the source tray 280, 1080 and the output tray 290, 1090 are secured within the system 200, 1000, the method 2000 continues to block 2004.

[0095] At block 2004, the items 205, 1005 within the source tray 280, 1080 are secured. In the example system 200, the items 205 can be secured by moving the paddle assembly 250 downward into the source tray 280 and optionally moving one or both of the paddles 252 inward, such that the items 205 are engaged between the paddles 252 and/or the paddle fingers 254. In the example system 1000, the items 1005 can be secured by moving the paddle assembly 1050 downward into the source tray 1080 and optionally moving paddle 1052b inward, such that the items 1005 are engaged between the paddle fingers 1054b and the detachable sidewall 1082. The paddle assembly 250, 1050, and/or paddles 252, 1052b can be moved by actuators such as actuators 1840-1880, based on instructions provided by the CPU 1820. After the items 205, 1005 within the source tray 280, 1080 are secured, the method 2000 continues to block 2006.

[0096] At block 2006, the items 205, 1005 are placed into a slidable configuration by moving the items 205, 1005, the item support platform 230, 1030, and/or a portion of the source tray 280, 1080. In the example system 200, the items 205 can be placed into a slidable configuration by simultaneously raising the lifting assembly 220 and the paddle assembly 250, followed by sliding the item support platform 230 to a position beneath the items 205, as shown in FIGS. 4-6. In the example system 1000, the items 1005 can be placed into a slidable configuration by disengaging the detachable sidewall 1082 of the source tray 1080 and sliding the item support platform 230 to a position adjacent to or partially beneath the source tray 1080, as shown in FIGS. 12-14. After the items 205, 1005 are placed into a slidable configuration, the method 2000 continues to block 2008.

[0097] At block 2008, the items 205, 1005 are positioned above the output tray 290, 1090 by horizontal translation (e.g., sliding). In the example system 200, the items 205 can be positioned above the output tray 290 by laterally translating the paddle assembly 250 and the item support platform 230 simultaneously, such that the items 205 are carried to a position in which they rest on the item support platform 230 and are disposed directly above the output tray 290. In the example system 1000, the items 1005 can be positioned above the output tray by laterally translating the paddle assembly 1050, including the detachable sidewall 1082 held by the grippers 1054a, such that the items 1005 slide from the source tray 1080 onto the item support platform 1030 and are disposed directly above the output tray 1090. After the items 205, 1005 are positioned above the output tray 290, 1090 by horizontal translation, the method 2000 continues to block 2010.

[0098] At block 2010, the item support platform 230, 1030 is translated to release the items 205, 1005 into the output tray 290, 1090. In the example system 200, the item support platform 230 may be translated to a disengaged position or to its previous position above the source tray 280, causing the items 205 to fall into the output tray 290 as the items 205 are retained laterally and the item support platform 230 is moved away. Similarly, in the example system 1000, the item support platform 1030 may be translated to a disengaged position to cause the items 1005 to fall into the output tray 1090 as the items 1005 are retained laterally and the item support platform 1030 is moved away. After the items 205, 1005 are released into the output tray 290, 1090, the method 2000 continues to block 2012.

[0099] The method 2000 terminates at block 2012, where the tray content transfer system 200, 1000 returns to its starting configuration. In the example system 200, the lifting assembly 220 moves downward to return to its initial position, the paddle assembly 250 moves upward and returns to its initial position above the source tray sub-frame 216, and the item support platform 230 returns to the disengaged position if not already in the disengaged position. In the example system 1000, the paddle assembly 1050 moves upward and returns to its initial position above the source tray sub-frame 216, and releases the detachable sidewall 1082 of the source tray 1080, and the item support platform 1030 returns to its initial disengaged position if not already in the disengaged position. After the tray content transfer system 200, 1000 returns to its starting

configuration, the method **2000** terminates. The system tray **280**, **1080** and the output tray **290**, **1090** can be removed from the system **200**, **1000** manually or by one or more automated systems such as the conveyors **217**, **219**, **1017**, and/or **1019**.

[0100] The foregoing description details certain embodiments of the systems, devices, and methods disclosed herein. It will be appreciated, however, that no matter how detailed the foregoing appears in text, the systems, devices, and methods can be practiced in many ways. It should be noted that the use of particular terminology when describing certain features or aspects of the invention should not be taken to imply that the terminology is being re-defined herein to be restricted to including any specific characteristics of the features or aspects of the technology with which that terminology is associated.

[0101] It will be appreciated by those skilled in the art that various modifications and changes may be made without departing from the scope of the described technology. Such modifications and changes are intended to fall within the scope of the embodiments. It will also be appreciated by those of skill in the art that parts included in one embodiment are interchangeable with other embodiments; one or more parts from a depicted embodiment can be included with other depicted embodiments in any combination. For example, any of the various components described herein and/or depicted in the Figures may be combined, interchanged, or excluded from other embodiments.

[0102] With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

[0103] It will be understood by those within the art that, in general, terms used herein are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms,

whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

[0104] All references cited herein are incorporated herein by reference in their entirety. To the extent publications and patents or patent applications incorporated by reference contradict the disclosure contained in the specification, the specification is intended to supersede and/or take precedence over any such contradictory material.

[0105] The term “comprising” as used herein is synonymous with “including,” “containing,” or “characterized by,” and is inclusive or open-ended and does not exclude additional, unrecited elements or method steps.

[0106] It is noted that some examples above may be described as a process, which is depicted as a flowchart, a flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel, or concurrently, and the process can be repeated. In addition, the order of the operations may be rearranged. A process is terminated when its operations are completed. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a software function, its termination corresponds to a return of the function to the calling function or the main function.

[0107] The above description discloses several methods and materials of the present invention. This invention is susceptible to modifications in the methods and materials, as well as alterations in the fabrication methods and equipment. Such modifications will become apparent to those skilled in the art from a consideration of this disclosure or practice of the invention disclosed herein. Consequently, it is not intended that this invention be limited to the specific embodiments disclosed herein, but that it cover all modifications and alternatives coming within the true scope and spirit of the invention as embodied in the attached claims.

Claims

1. A method for transferring items from a tray, the method comprising: receiving a first tray containing one or more items at a first location in a tray content transfer system; receiving a second tray in the tray content transfer system at a second location spaced along a longitudinal axis from the first location; securing the one or more items relative to the first tray; moving an item support platform parallel to the longitudinal axis; moving the one or more items such that the one or more items rest on the item support platform above the second tray; and moving the item support platform to release the one or more items into the second tray.
2. The method of claim 1, wherein securing the one or more items comprises: moving a paddle assembly downward into the first tray such that the one or more items are disposed between a plurality of paddles of the paddle assembly; and engaging the one or more items between the plurality of paddles, wherein the one or more items remain engaged between the plurality of paddles until the one or more items are released into the second tray.
3. The method of claim 1, wherein moving the one or more items such that the one or more items rest on the item support platform above the second tray comprises contacting an end one of the one or more items by a paddle; and pushing, by the paddle, the one or more items onto the item support platform.
4. The method of claim 2, further comprising moving the paddle assembly upward to a raised position in which the items are disposed higher than a top of the first tray.
5. The method of claim 4, wherein moving the item support platform parallel to the longitudinal axis comprises moving the item support platform to a position between the paddle assembly and the top of the first tray.
6. The method of claim 1, further comprising detaching a detachable sidewall from the first tray.

7. The method of claim 6, wherein detaching the detachable sidewall comprises moving a first paddle to detach the detachable sidewall from the first tray.
 8. The method of claim 1, further comprising lifting the one or more items out of the first tray by a lifting assembly.
 9. The method of claim 1, further comprising: detecting the presence of the first tray in the tray content transfer system; and initiating, by a controller, securing the one or more items relative to the first tray.
 10. The method of claim 1, wherein moving the item support platform to release the one or more items comprises moving the item support platform parallel to the longitudinal axis.
 11. The method of claim 1, wherein the item support platform comprises at least two sections, each of the at least two sections being rotatable about individual rotation axes.
 12. The method of claim 11, wherein moving the item support platform to release the one or more items comprises moving a first section of the item support platform away from a second section of the item support platform.
 13. The method of claim 8, wherein moving the item support platform to release the one or more items comprises rotating at least one of the at least two sections about the individual rotation axis.
 14. The method of claim 1, wherein moving an item support platform parallel to the longitudinal axis comprises moving the item support platform to a first position above the tray receiving area.
 15. The method of claim 14, further comprising contacting a paddle on an end item of the one or more items; and moving the paddle to move the stack of items out of the first tray.
 16. The method of claim 15, wherein moving the one or more items along the longitudinal axis such that the one or more items rest on the item support platform above the second tray comprises moving the item support platform to a second position proximate the second tray.
 17. The method of claim 1, further comprising lifting, via a lifting assembly at the first location, the one or more items in the first tray in a vertical axis, the vertical axis being generally perpendicular to the longitudinal axis.
 18. The method of claim 17, wherein the lifting assembly comprises a plurality of fingers, and wherein lifting the one or more items in the first tray comprises contacting the one or more items by the plurality of fingers through one or more openings in the bottom of the first tray.
 19. The method of claim 17, wherein the lifting assembly is configured to move a bottom of the first tray along the vertical axis thereby moving the one or more items along the vertical axis.
 20. A tray unloader comprising: means for receiving a first tray containing one or more items at a first location in a tray content transfer system; means for receiving a second tray in the tray content transfer system at a second location spaced along a longitudinal axis from the first location; means for securing the one or more items relative to the first tray; means for moving an item support platform parallel to the longitudinal axis; means for moving the one or more items such that the one or more items rest on the item support platform above the second tray; and means for moving the item support platform to release the one or more items into the second tray.
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